



AI Systems Engineering

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Projects Guidelines

Types of Projects

Innovation-driven

- Identify a Business or Societal need
- Find an application/task of interest and explore how to approach/solve it effectively, often with existing models
- Devise and develop a use case to solve a real-world/realistic problem, with a final proof of concept. Development-oriented projects
- **Teacher support:** for eliciting and defining requirements
- Follow the steps in the next slides

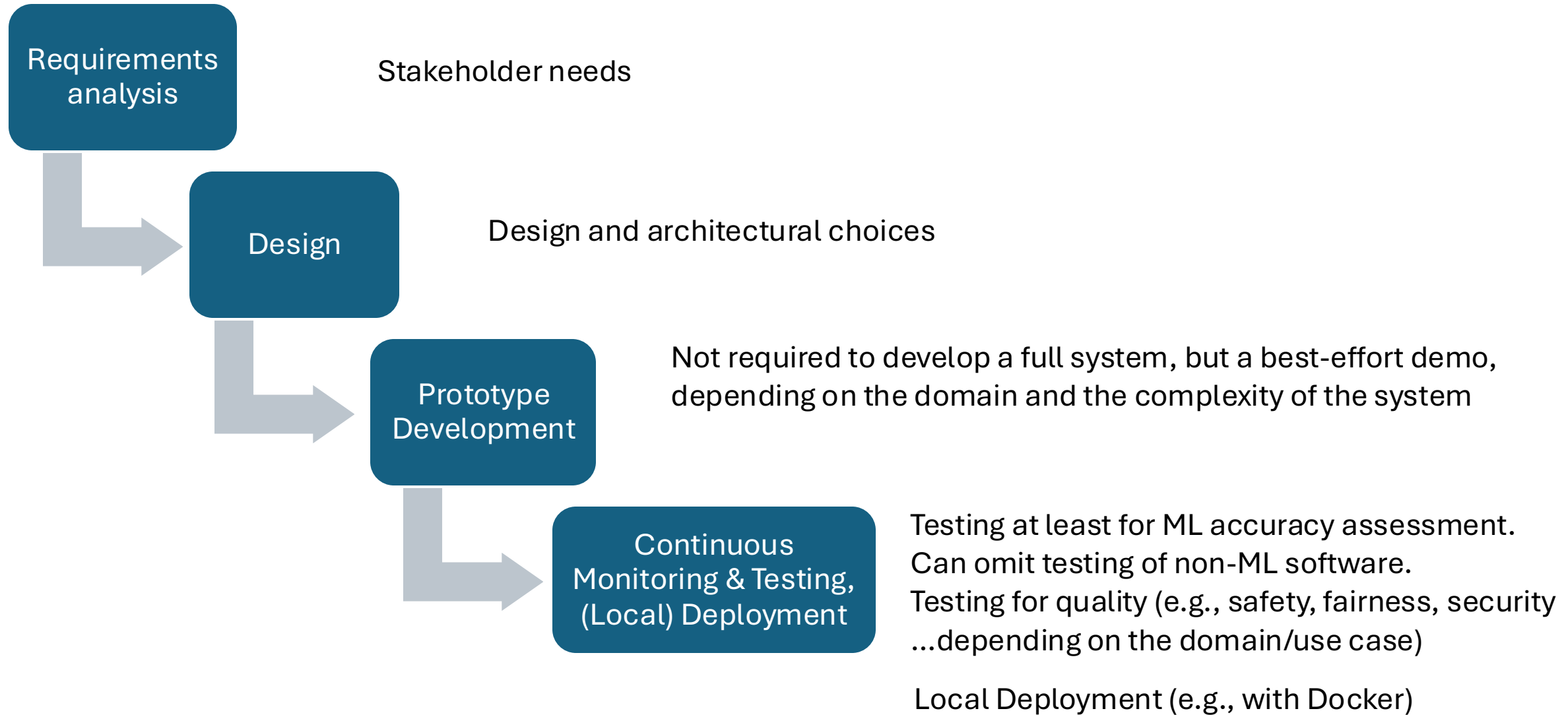
Research-driven

- Identify a scientific unsolved problem/challenges or solved problems needing better solutions
- Explore existing solutions and propose (with the help of the teacher) a new one
- More oriented to trustworthiness/quality evaluation (with plenty of open problems)
- **Teacher support:** for defining the “research” idea/hypothesis and for planning the DoE
- Follow the steps in the next slides

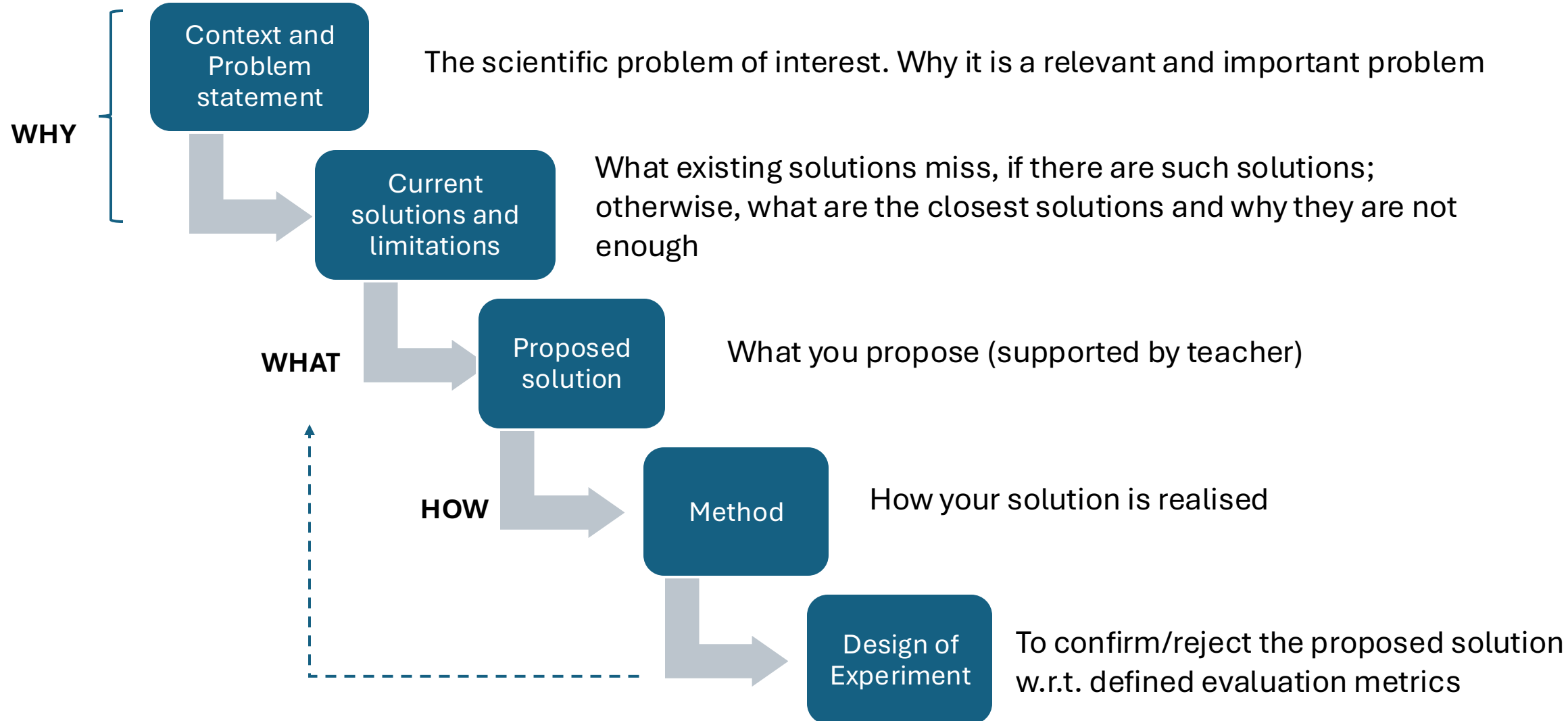
Both can be thought completely from scratch or starting from the next examples (see later)

In both cases, follow along with the teacher to precisely set the goals before starting

Innovation-driven: minimal steps



Research-driven minimal steps



Example of Projects

- Any of the below projects can be tackled as a Innovation-driven (INN) or Research-driven (RES), depending on the goal and the specific use case (to discuss and be agreed with teacher depending on the group's interest)
 - I give a tentative tag INN or RES, but are not strict: any project can be defined as INN or RES
- Multi-group-cooperation: more groups can cooperate on a more complex project: for instance, one group develops a use case (INN) while another group evaluates it with existing techniques (INN) or with new techniques (RES)
- Examples
 - Consumer goods and services **data analytics** (marketing, advertising, manufacturing, e-commerce)
 - Data analytics, e.g., ML for sales prediction, consumer profiling, e-commerce, market analysis (INN)
 - LLM-Powered **Virtual Assistants (Human-Machine Cooperation)**, such as:
 - **Virtual Assistants for human-machine cooperation in sensitive domains.** For instance, a virtual teacher for schools/education: train an LLM to assist students with personalized plans (e.g., considering the scores of the student in the past in various subjects and personal features) (INN). Test it for fairness (e.g., such an LLM could discriminate among students with different background, ethnicity, economic status, ...), privacy, toxicity,... (RES).
 - **Virtual Agent for Software Engineering:** Assist engineers to generate requirements, design, test cases, code analysis, attack scenarios and/or unsafe scenarios in any domain of interest (INN and/or RES)
 - ...

Example of Projects

- ...
- **Virtual Doctor:** train an LLM to interact with a doctor to help diagnosis (INN). Need to test the answers for (health) safety (it can give wrong diagnosis) as well as for fairness (it can discriminate) – empirical studies can be run on this to check alternative solutions to raise accuracy (RES)
- **Misinformation detector:** train an LLM-based system to score news/posts for fake content (INN/RES)
- **Virtual Hacker (or fakes generator):** train an LLM to generate security/privacy attacks (or fake news), useful to design defenses (to test fake news detectors) (INN and/or RES)
- **Virtual legal assistant** to support arguments generation, arguments mining, claim detection (INN/RES)
- **Chatbots** for customer service, same as above (INN and/or RES)
- **Scientific discovery:** LLM-based system for scientific hypothesis generation (e.g., in biology, health, genomics)
- *Can easily envision other such “assistants” or agents...*
- **Evaluation** (and protection against) any **virtual LLM-powered entity/agents** for **unsafe, unsecure, unfair** behaviours (e.g., when interacting with protected categories: children/young, disables, ...) (RES)
- **Testing, Evaluation, Risk Assessment/Mitigation of LLM-Systems** (includes many of the above cases) as well as on classical ML systems (like the next ones) (RES)
 - Many open challenge, problems to solve (hallucination, privacy, fairness, toxicity, robustness to attacks, ...)

Example of Projects

- **AIoT and hardware/software systems**

- Smart buildings, smart homes, smart cities (INN)
 - E.g., gather data about energy consumption, detect anomalous consumptions, predict consumptions, optimize energy distribution (Energy Management system, power grids) ...
- Against **climate change**, such as (under)water or air monitoring (and detection/prediction) against pollution, infrastructure monitoring (e.g., dams, bridges, ...)
 - E.g., to prevent natural disasters, such as monitoring/detecting/predicting volcanoes, earthquakes,...
- Agriculture, e.g., **precision farming**, for personalized treatments based on sensors measurement (INN)
- **Well-being** (INN)
 - E.g., personal trainers or nutritionists working on data from wearable sensors, or user-inputted data (e.g., meals, weight,...) to suggest training plans

Example of Projects

- **Autonomous Driving Systems (ADS) or Advanced Driver Assistance Systems (ADAS) or Unmanned Aerial Vehicles (UAV)**
 - Testing for safety assessment and generation of security attacks ([RES](#))
 - Simulation-based system-level testing ([RES](#))
 - Traffic and Route planning (gather data, and characterise/predict traffic patterns, ...) ([INN](#))
- **Banking/Finance**, e.g., for granting loans, predicting stock price, fraud detection ([INN](#)).
Testing for fairness and privacy ([RES](#))
- For **IT infrastructure** such as cloud, clusters (failure/log analysis, anomaly/error detection, diagnosis, failure prediction, ..) ([INN](#) and/or [RES](#))

Example of Projects

- Other domains...
 - **Health care**
 - Medical diagnosis from images or from unstructured data (e.g., from textual or voice data for mental/psychological safety, anxiety/depression) ([INN](#)), reliab. eval. of diagnosis ([RES](#))
 - Testing other AI systems against cognitive manipulation behaviors (mental safety), e.g., for protecting children and young ([RES](#))
 - **Robotics**
 - E.g. for healthcare (e.g., gather data about patients and alerting on anomalies detection)
 - For process automation in Industry 4.0
 - **Computer vision**, object detection, surveillance
 - ...
- *Note: can combine multiple ideas (e.g., LLM used in an ADS context,...)*

Rules

- Number of members
 - No limit, depending on the project complexity
- Multi-team Cooperation (see the note in the previous slides)
- Delivery to the teacher via a GitHub Repository
- Hints:
 - Use Simulation/Emulation (for coping with missing Data and/or Hardware)
 - Use Office Hours and messages on Team to interact with the teacher

Evaluation Criteria

- Impact, relevance, significance
- Correctness, rigor, soundness
- Originality
- Presentation
- Reproducibility, replicability